

UNIVERSIDAD DE GUADALAJARA

CENTRO UNIVERSITARIO DE LOS VALLES



Maestría en Ingeniería de Software

Course: Administración de la configuración del software

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Project Proposal v2

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PROJECT PROPOSAL

StudyGo — Mobile app for gamified homework

1. THE COMPANY

Name: Blue Mobile Solutions S.A. de C.V.

Location: Guadalajara, Jalisco.

Years in business: 7 years.

Size: 24 employees.

Services: custom mobile app development for Android and iOS.

2. THE CLIENT

Name: Amber High School

School level: high school (general program, 6 semesters)

Type: private school

Location: Guadalajara, Jalisco

Project contact: academic office

3. PROBLEM STATEMENT

The teachers at Amber High School want a mobile app that is easy to use for their students, because all of them have a cell phone but not all of them have a computer.

In the app, each teacher can assign homework for the subject they teach, and the student does it on their phone. The homework is gamified, so the student feels more motivated to do it and to finish all of it.

Teachers can see the scores of each student and of the whole group. This helps them find which topics they need to review again and which ones the students understand better.

The student can see their own progress in each subject, and parents can see the progress and the learning level of their children.

4. PROPOSED SOLUTION

We propose to develop StudyGo, a mobile app for Amber High School.

In the app, the teacher assigns homework for the subject they teach. The student does it on their cell phone. The homework is gamified: the student earns points when they finish it and can see their progress in each subject.

The teacher can see the statistics of their subject. There they see the scores of each student and of the whole group. With this information they find which topics need more review and which ones are better understood.

The parent role is read only. It shows the same information the student sees, but it does not allow doing the homework.

App roles

The app has 3 user roles:

1. Teacher
2. Student
3. Parent

Technology

The client did not say which operating system the teachers, the students and the parents use on their phones. For this reason, the app will be built with Flutter, which is a cross-platform technology.

With Flutter we write one code and from it we build two versions, one for Android and one for iOS. This has two benefits: all users can use the app no matter what phone they have, and the work costs less than making two separate apps.

5. HOW THE APP WORKS

The teacher role allows creating homework, marking the subject and the topic it belongs to. The app has three types of homework: multiple choice, true or false, and filling in a sentence by dragging words. All three are graded by the app.

Gamification means using game elements inside an app that is not a game. It helps the user feel motivated to use it. In this app it is used to motivate the student to do their homework. The student earns points and badges for the homework they finish, goes up a level in each subject, and appears in the ranking of their group.

6. MODULES

The system has 8 modules:

1. Users and login
2. Subjects and groups
3. Homework
4. Points and badges
5. Student progress
6. Teacher reports
7. Parent view
8. Notices

7. SCOPE

The project includes the 8 modules listed in this document, in their version for Android and iOS.

Any other feature that is not on that list is outside this proposal and will be planned for a second version of the system.

8. ESTIMATED TIME

The development takes 5 months. We add 1 more month as extra time for problems, so the project is delivered in 6 months.

The team works with the Scrum method. The work is divided into 10 sprints of 2 weeks each. In every sprint the team agrees to finish one part of the system, and at the end of the sprint the school is shown what is working.

How the work is divided into sprints

SPRINTS	WORK
1 and 2	Requirements analysis and screen design
3	Module 1. Users and login
4	Module 2. Subjects and groups
5	Module 3. Homework: create and assign
6	Module 3. Homework: answer the three types
7	Modules 4 and 5. Points and badges, student progress
8	Modules 6 and 7. Teacher reports, parent view
9	Module 8. Notices
10	Final testing and delivery

9. PROJECT TEAM

For this project we assigned 8 people: 3 programmers and 5 people in the other roles.

ROLE	SKILLS
Project manager	Scrum, planning, Jira
Requirements analyst	Interviews, user stories, UML
UI/UX designer	Figma, Material Design, usability testing
Programmer 1 (technical lead, senior)	Flutter, Dart, Firebase, Git
Programmer 2 (junior)	Flutter, Dart, SQLite, REST services
Programmer 3 (backend, semi-senior)	Node.js, NestJS, PostgreSQL, Docker
QA engineer	Test cases, Postman, Appium
Infrastructure engineer	Docker, cloud services, Linux

10. BUDGET

Team salaries

ROLE	MONTHLY SALARY	MONTHS	TOTAL
Programmer 1 (senior)	\$42,000	6	\$252,000
Programmer 2 (junior)	\$20,000	5	\$100,000
Programmer 3 (semi-senior)	\$30,000	6	\$180,000
Project manager	\$40,000	3	\$120,000
Requirements analyst	\$26,000	2	\$52,000
UI/UX designer	\$24,000	2	\$48,000
QA engineer	\$22,000	3	\$66,000
Infrastructure engineer	\$30,000	2	\$60,000
Salaries subtotal			\$878,000

Other expenses

CONCEPT	AMOUNT
Office rent, the part used by the team ($\$7,000 \times 6$ months)	\$42,000
Internet and utilities ($\$2,000 \times 6$ months)	\$12,000
Test phones	\$25,000
Licenses and cloud services during development	\$29,500
Subtotal	\$108,500

Total cost of the project

CONCEPT	AMOUNT
Salaries subtotal	\$878,000
Other expenses	\$108,500
Company profit (25%)	\$246,625
Total	\$1,233,125 MXN